

# Phase I/II Operating Characteristics

Method comparison - N = 30 (AIDE is an alpha-based model)

|                            | Dose 1 | Dose 2 | Dose 3 | Dose 4 | Dose 5 | Stop % |
|----------------------------|--------|--------|--------|--------|--------|--------|
| Scenario 1                 |        |        |        |        |        |        |
| DLT rate                   | 0.3    | 0.35   | 0.4    | 0.45   | 0.5    |        |
| Efficacy rate              | 0.05   | 0.06   | 0.07   | 0.08   | 0.09   |        |
| Utility 2                  | -0.2   | -0.3   | -0.3   | -0.4   | -0.4   |        |
| Utility 3                  | 31     | 29.6   | 28.2   | 26.8   | 25.4   |        |
| OBD selection (%)          |        |        |        |        |        |        |
| BOIN12                     | 36.1   | 14.4   | 5.9    | 1.5    | 0.7    | 41.4   |
| U-BOIN                     | 27.8   | 20.4   | 16     | 13.4   | 9.5    | 12.9   |
| EffTox                     | 5      | 4      | 6      | 3      | 6      | 75     |
| AIDE one_stage (Utility 2) | 35.3   | 16.8   | 2.4    | 0.6    | 0.2    | 44.7   |
| AIDE one_stage (Utility 3) | 35.4   | 15.6   | 2      | 0      | 0.1    | 46.9   |
| AIDE two_stage (Utility 2) | 31.3   | 10.3   | 1.5    | 0.5    | 0.2    | 56.2   |
| AIDE two_stage (Utility 3) | 33.6   | 9      | 0.6    | 0.4    | 0.2    | 56.2   |
| Mean patients treated      |        |        |        |        |        |        |
| BOIN12                     | 13.5   | 6      | 3.2    | 1.9    | 0.9    |        |
| U-BOIN                     | 9.8    | 5.4    | 4.3    | 3.8    | 3.6    |        |
| EffTox                     | 8      | 3.2    | 3      | 2      | 2      |        |
| AIDE one_stage (Utility 2) | 14.6   | 6.3    | 2.5    | 1.4    | 0.4    |        |
| AIDE one_stage (Utility 3) | 15.1   | 5.9    | 2.3    | 1.4    | 0.4    |        |
| AIDE two_stage (Utility 2) | 15.4   | 3.9    | 2.1    | 0.9    | 0.4    |        |
| AIDE two_stage (Utility 3) | 15.3   | 3.9    | 2.1    | 0.9    | 0.4    |        |
| Scenario 2                 |        |        |        |        |        |        |
| DLT rate                   | 0.15   | 0.2    | 0.3    | 0.35   | 0.45   |        |
| Efficacy rate              | 0.2    | 0.25   | 0.5    | 0.5    | 0.5    |        |
| Utility 2                  | 0.1    | 0.1    | 0.2    | 0.1    | 0.1    |        |
| Utility 3                  | 46     | 47     | 58     | 56     | 52     |        |
| OBD selection (%)          |        |        |        |        |        |        |
| BOIN12                     | 29.2   | 24.7   | 33.9   | 9.2    | 1.5    | 1.5    |
| U-BOIN                     | 11.2   | 11.5   | 40.6   | 24.1   | 11.7   | 0.9    |
| EffTox                     | 72     | 7      | 5      | 6      | 9      | 1      |
| AIDE one_stage (Utility 2) | 25.2   | 18.3   | 39.3   | 13.9   | 2.3    | 1      |
| AIDE one_stage (Utility 3) | 30.8   | 19.1   | 35.4   | 11.9   | 1.8    | 1      |
| AIDE two_stage (Utility 2) | 34.9   | 16     | 31.9   | 11.6   | 3.8    | 1.8    |
| AIDE two_stage (Utility 3) | 38.3   | 16.1   | 30.8   | 10.1   | 2.9    | 1.8    |
| Mean patients treated      |        |        |        |        |        |        |
| BOIN12                     | 8.6    | 7.8    | 8.6    | 3.6    | 1.1    |        |
| U-BOIN                     | 8.7    | 6.9    | 8.2    | 4.3    | 1.8    |        |
| EffTox                     | 22.6   | 2.5    | 1.6    | 1.2    | 1.7    |        |
| AIDE one_stage (Utility 2) | 9.2    | 7      | 9      | 3.8    | 0.9    |        |
| AIDE one_stage (Utility 3) | 9.8    | 7.2    | 8.7    | 3.5    | 0.7    |        |
| AIDE two_stage (Utility 2) | 11.4   | 5.1    | 7.2    | 3.9    | 2.2    |        |
| AIDE two_stage (Utility 3) | 11.4   | 5.1    | 7.2    | 3.9    | 2.2    |        |
| Scenario 3                 |        |        |        |        |        |        |
| DLT rate                   | 0.15   | 0.2    | 0.3    | 0.35   | 0.45   |        |
| Efficacy rate              | 0.1    | 0.4    | 0.4    | 0.4    | 0.4    |        |
| Utility 2                  | -0.1   | 0.2    | 0.1    | 0.1    | -0.1   |        |
| Utility 3                  | 40     | 56     | 52     | 50     | 46     |        |
| OBD selection (%)          |        |        |        |        |        |        |
| BOIN12                     | 13.2   | 56     | 20.6   | 6.9    | 1.2    | 2.1    |
| U-BOIN                     | 3.8    | 44.7   | 24.5   | 16.2   | 10     | 0.8    |
| EffTox                     | 49     | 19     | 8      | 7      | 15     | 2      |
| AIDE one_stage (Utility 2) | 9.6    | 46.7   | 26.9   | 12.3   | 2.1    | 2.4    |
| AIDE one_stage (Utility 3) | 12.4   | 53.6   | 21.1   | 8.7    | 1.7    | 2.5    |
| AIDE two_stage (Utility 2) | 17.5   | 40.4   | 21.1   | 8.2    | 2.8    | 10     |
| AIDE two_stage (Utility 3) | 20.2   | 41.4   | 20.9   | 5.4    | 2.1    | 10     |
| Mean patients treated      |        |        |        |        |        |        |
| BOIN12                     | 6.9    | 11.3   | 6.4    | 3.5    | 1.6    |        |
| U-BOIN                     | 7.2    | 10.7   | 6.7    | 3.6    | 1.7    |        |
| EffTox                     | 18.8   | 4.8    | 2.6    | 1.5    | 1.9    |        |
| AIDE one_stage (Utility 2) | 6.6    | 11.3   | 6.8    | 3.8    | 1.1    |        |
| AIDE one_stage (Utility 3) | 6.8    | 12.2   | 6.4    | 3.4    | 0.9    |        |
| AIDE two_stage (Utility 2) | 9.7    | 7.5    | 6.1    | 3.6    | 2      |        |
| AIDE two_stage (Utility 3) | 9.7    | 7.4    | 6.2    | 3.6    | 2      |        |

# Phase I/II Operating Characteristics

Method comparison - N = 30 (AIDE is an alpha-based model)

|                            | Dose 1 | Dose 2 | Dose 3 | Dose 4 | Dose 5 | Stop % |
|----------------------------|--------|--------|--------|--------|--------|--------|
| Scenario 4                 |        |        |        |        |        |        |
| DLT rate                   | 0.05   | 0.1    | 0.18   | 0.3    | 0.4    |        |
| Efficacy rate              | 0.05   | 0.1    | 0.25   | 0.5    | 0.55   |        |
| Utility 2                  | 0      | 0      | 0.1    | 0.2    | 0.1    |        |
| Utility 3                  | 41     | 42     | 47.8   | 58     | 57     |        |
| OBD selection (%)          |        |        |        |        |        |        |
| BOIN12                     | 15.6   | 14.2   | 25.6   | 36.9   | 7.4    | 0.3    |
| U-BOIN                     | 3.7    | 6      | 18.4   | 43.5   | 28.4   | 0      |
| EffTox                     | 7      | 14     | 20     | 17     | 40     | 1      |
| AIDE one_stage (Utility 2) | 6.7    | 6.6    | 17.5   | 52     | 17     | 0.2    |
| AIDE one_stage (Utility 3) | 9.6    | 7.8    | 16.9   | 52.1   | 13.4   | 0.2    |
| AIDE two_stage (Utility 2) | 6.8    | 7.4    | 21.4   | 42.6   | 19.9   | 1.9    |
| AIDE two_stage (Utility 3) | 11.5   | 9      | 19.1   | 40.8   | 17.7   | 1.9    |
| Mean patients treated      |        |        |        |        |        |        |
| BOIN12                     | 5.4    | 5.9    | 6.9    | 8.7    | 3.1    |        |
| U-BOIN                     | 5.1    | 6.1    | 7.4    | 7.7    | 3.9    |        |
| EffTox                     | 9.4    | 4.8    | 5.3    | 3.7    | 6.4    |        |
| AIDE one_stage (Utility 2) | 4.2    | 4.6    | 6.7    | 10.6   | 3.9    |        |
| AIDE one_stage (Utility 3) | 4.5    | 4.6    | 6.8    | 10.7   | 3.4    |        |
| AIDE two_stage (Utility 2) | 4.5    | 4.2    | 6.5    | 8      | 6.5    |        |
| AIDE two_stage (Utility 3) | 4.5    | 4.2    | 6.5    | 7.9    | 6.6    |        |
| Scenario 5                 |        |        |        |        |        |        |
| DLT rate                   | 0.07   | 0.12   | 0.17   | 0.22   | 0.3    |        |
| Efficacy rate              | 0.05   | 0.3    | 0.3    | 0.3    | 0.3    |        |
| Utility 2                  | 0      | 0.2    | 0.1    | 0.1    | 0      |        |
| Utility 3                  | 40.2   | 53.2   | 51.2   | 49.2   | 46     |        |
| OBD selection (%)          |        |        |        |        |        |        |
| BOIN12                     | 7      | 42.3   | 25.2   | 18     | 7      | 0.5    |
| U-BOIN                     | 2.9    | 37.5   | 26.5   | 21.3   | 11.7   | 0.1    |
| EffTox                     | 22     | 22     | 17     | 13     | 25     | 2      |
| AIDE one_stage (Utility 2) | 5.5    | 38.4   | 25.4   | 19.7   | 10.7   | 0.3    |
| AIDE one_stage (Utility 3) | 8.1    | 43.6   | 24     | 17.6   | 6.4    | 0.3    |
| AIDE two_stage (Utility 2) | 4.9    | 35.3   | 26.3   | 18.1   | 11.2   | 4.2    |
| AIDE two_stage (Utility 3) | 7.8    | 41.1   | 23.4   | 16.3   | 7.2    | 4.2    |
| Mean patients treated      |        |        |        |        |        |        |
| BOIN12                     | 4.5    | 7.8    | 6.7    | 6.7    | 4.4    |        |
| U-BOIN                     | 5.2    | 8.5    | 6.9    | 5.7    | 3.9    |        |
| EffTox                     | 12.8   | 5.9    | 4.7    | 2.8    | 3.5    |        |
| AIDE one_stage (Utility 2) | 3.9    | 7.3    | 7      | 7.1    | 4.7    |        |
| AIDE one_stage (Utility 3) | 4      | 7.9    | 7.3    | 6.9    | 3.9    |        |
| AIDE two_stage (Utility 2) | 5      | 6.4    | 6.7    | 5.7    | 5.7    |        |
| AIDE two_stage (Utility 3) | 5      | 6.4    | 6.7    | 5.7    | 5.6    |        |
| Scenario 6                 |        |        |        |        |        |        |
| DLT rate                   | 0.29   | 0.34   | 0.38   | 0.43   | 0.52   |        |
| Efficacy rate              | 0.3    | 0.32   | 0.35   | 0.38   | 0.4    |        |
| Utility 2                  | 0      | 0      | 0      | -0.1   | -0.1   |        |
| Utility 3                  | 46.4   | 45.6   | 45.8   | 45.6   | 43.2   |        |
| OBD selection (%)          |        |        |        |        |        |        |
| BOIN12                     | 64.3   | 16.7   | 5.3    | 1.7    | 0.4    | 11.6   |
| U-BOIN                     | 25.6   | 17.2   | 13.9   | 17.6   | 14.3   | 11.4   |
| EffTox                     | 85     | 1      | 2      | 1      | 1      | 10     |
| AIDE one_stage (Utility 2) | 63.5   | 17.5   | 4.9    | 1.9    | 0.5    | 11.7   |
| AIDE one_stage (Utility 3) | 66.7   | 16.2   | 3.7    | 1.2    | 0.5    | 11.7   |
| AIDE two_stage (Utility 2) | 67.6   | 14.2   | 3.2    | 1.2    | 0.5    | 13.3   |
| AIDE two_stage (Utility 3) | 68.7   | 13.3   | 3.1    | 1      | 0.4    | 13.5   |
| Mean patients treated      |        |        |        |        |        |        |
| BOIN12                     | 15.9   | 6.4    | 3.3    | 1.8    | 0.6    |        |
| U-BOIN                     | 11.9   | 6.1    | 4.3    | 3.1    | 1.9    |        |
| EffTox                     | 25.6   | 1.1    | 0.6    | 0.3    | 0.3    |        |
| AIDE one_stage (Utility 2) | 17.5   | 6.1    | 2.7    | 1.3    | 0.3    |        |
| AIDE one_stage (Utility 3) | 18.1   | 6      | 2.4    | 1.2    | 0.3    |        |
| AIDE two_stage (Utility 2) | 19.6   | 3.9    | 2.5    | 1.1    | 0.4    |        |
| AIDE two_stage (Utility 3) | 19.6   | 3.9    | 2.5    | 1.1    | 0.4    |        |

# Phase I/II Operating Characteristics

Method comparison - N = 30 (AIDE is an alpha-based model)

|                            | Dose 1 | Dose 2 | Dose 3 | Dose 4 | Dose 5 | Stop % |
|----------------------------|--------|--------|--------|--------|--------|--------|
| Scenario 7                 |        |        |        |        |        |        |
| DLT rate                   | 0.16   | 0.2    | 0.26   | 0.31   | 0.36   |        |
| Efficacy rate              | 0.2    | 0.5    | 0.5    | 0.5    | 0.5    |        |
| Utility 2                  | 0      | 0.3    | 0.2    | 0.2    | 0.1    |        |
| Utility 3                  | 45.6   | 62     | 59.6   | 57.6   | 55.6   |        |
| OBD selection (%)          |        |        |        |        |        |        |
| BOIN12                     | 12.8   | 51.1   | 20.7   | 9.8    | 3.4    | 2.2    |
| U-BOIN                     | 4.3    | 48.3   | 25.8   | 13.7   | 6.2    | 1.7    |
| EffTox                     | 72     | 8      | 6      | 4      | 10     | 1      |
| AIDE one_stage (Utility 2) | 13.1   | 48.1   | 22.8   | 11.5   | 3.5    | 1      |
| AIDE one_stage (Utility 3) | 15.5   | 52.7   | 20     | 9.1    | 1.7    | 1      |
| AIDE two_stage (Utility 2) | 26.4   | 36.6   | 20.5   | 10.6   | 3.7    | 2.2    |
| AIDE two_stage (Utility 3) | 28.2   | 39     | 18.9   | 8.6    | 3      | 2.3    |
| Mean patients treated      |        |        |        |        |        |        |
| BOIN12                     | 6.9    | 12.5   | 5.9    | 3.1    | 1.2    |        |
| U-BOIN                     | 7.6    | 10.7   | 6.5    | 3.4    | 1.4    |        |
| EffTox                     | 22.9   | 2.8    | 1.6    | 1.1    | 1.4    |        |
| AIDE one_stage (Utility 2) | 7.6    | 12.6   | 5.8    | 2.9    | 0.9    |        |
| AIDE one_stage (Utility 3) | 8      | 13.5   | 5.3    | 2.5    | 0.6    |        |
| AIDE two_stage (Utility 2) | 10.8   | 7      | 5.6    | 3.8    | 2.5    |        |
| AIDE two_stage (Utility 3) | 10.8   | 6.9    | 5.7    | 3.7    | 2.6    |        |
| Scenario 8                 |        |        |        |        |        |        |
| DLT rate                   | 0.32   | 0.37   | 0.42   | 0.48   | 0.52   |        |
| Efficacy rate              | 0.1    | 0.15   | 0.2    | 0.3    | 0.4    |        |
| Utility 2                  | -0.2   | -0.2   | -0.2   | -0.2   | -0.1   |        |
| Utility 3                  | 33.2   | 34.2   | 35.2   | 38.8   | 43.2   |        |
| OBD selection (%)          |        |        |        |        |        |        |
| BOIN12                     | 46.2   | 15     | 5.1    | 1.6    | 0.5    | 31.6   |
| U-BOIN                     | 14.8   | 10.2   | 14.5   | 17.4   | 26.7   | 16.4   |
| EffTox                     | 27     | 11     | 8      | 5      | 9      | 39     |
| AIDE one_stage (Utility 2) | 44.6   | 16.2   | 3.1    | 0.9    | 0.3    | 34.9   |
| AIDE one_stage (Utility 3) | 45.9   | 15.4   | 2.3    | 0.9    | 0.3    | 35.2   |
| AIDE two_stage (Utility 2) | 43.2   | 9.2    | 2.8    | 1.2    | 0.3    | 43.3   |
| AIDE two_stage (Utility 3) | 45     | 8.7    | 2      | 0.9    | 0.2    | 43.2   |
| Mean patients treated      |        |        |        |        |        |        |
| BOIN12                     | 14.5   | 5.9    | 3.1    | 1.7    | 0.7    |        |
| U-BOIN                     | 8.9    | 4.7    | 4.2    | 4.3    | 4.1    |        |
| EffTox                     | 14.3   | 3.3    | 2.4    | 1.5    | 1.9    |        |
| AIDE one_stage (Utility 2) | 15.7   | 6.2    | 2.3    | 1.2    | 0.3    |        |
| AIDE one_stage (Utility 3) | 16     | 6      | 2.2    | 1.1    | 0.3    |        |
| AIDE two_stage (Utility 2) | 17.3   | 3.7    | 1.9    | 0.8    | 0.3    |        |
| AIDE two_stage (Utility 3) | 17.4   | 3.7    | 1.9    | 0.8    | 0.2    |        |
| Scenario 9                 |        |        |        |        |        |        |
| DLT rate                   | 0.15   | 0.18   | 0.22   | 0.26   | 0.29   |        |
| Efficacy rate              | 0.05   | 0.08   | 0.2    | 0.25   | 0.5    |        |
| Utility 2                  | -0.1   | -0.1   | 0      | 0      | 0.2    |        |
| Utility 3                  | 37     | 37.6   | 43.2   | 44.6   | 58.4   |        |
| OBD selection (%)          |        |        |        |        |        |        |
| BOIN12                     | 18.9   | 16.2   | 22.4   | 12.6   | 25.3   | 4.6    |
| U-BOIN                     | 3.8    | 3.9    | 13.8   | 19.1   | 58     | 1.4    |
| EffTox                     | 8      | 8      | 12     | 8      | 56     | 9      |
| AIDE one_stage (Utility 2) | 14.7   | 15.3   | 22.3   | 16.8   | 26.5   | 4.4    |
| AIDE one_stage (Utility 3) | 21.7   | 16.8   | 18.6   | 15.2   | 22.9   | 4.8    |
| AIDE two_stage (Utility 2) | 13.4   | 10.5   | 17.5   | 13.2   | 26.8   | 18.6   |
| AIDE two_stage (Utility 3) | 20.9   | 10.6   | 14.8   | 12     | 23     | 18.7   |
| Mean patients treated      |        |        |        |        |        |        |
| BOIN12                     | 6.6    | 5.9    | 5.7    | 5.2    | 6.2    |        |
| U-BOIN                     | 6.8    | 5.6    | 6.3    | 5.2    | 6      |        |
| EffTox                     | 9.3    | 3.9    | 4.6    | 3      | 7.7    |        |
| AIDE one_stage (Utility 2) | 7      | 6.1    | 5.9    | 5.5    | 5.1    |        |
| AIDE one_stage (Utility 3) | 7.6    | 6      | 5.8    | 5.4    | 4.8    |        |
| AIDE two_stage (Utility 2) | 8.9    | 4.3    | 5.3    | 4      | 5.3    |        |
| AIDE two_stage (Utility 3) | 8.9    | 4.3    | 5.3    | 4      | 5.3    |        |

# Phase I/II Operating Characteristics

Method comparison - N = 30 (AIDE is an alpha-based model)

|                            | Dose 1 | Dose 2 | Dose 3 | Dose 4 | Dose 5 | Stop % |
|----------------------------|--------|--------|--------|--------|--------|--------|
| Scenario 10                |        |        |        |        |        |        |
| DLT rate                   | 0.23   | 0.28   | 0.32   | 0.39   | 0.43   |        |
| Efficacy rate              | 0.05   | 0.4    | 0.42   | 0.45   | 0.5    |        |
| Utility 2                  | -0.2   | 0.1    | 0.1    | 0.1    | 0.1    |        |
| Utility 3                  | 33.8   | 52.8   | 52.4   | 51.4   | 52.8   |        |
| OBD selection (%)          |        |        |        |        |        |        |
| BOIN12                     | 14.6   | 43.9   | 16.1   | 4      | 2.2    | 19.2   |
| U-BOIN                     | 1.2    | 32.8   | 27.1   | 17     | 17.3   | 4.6    |
| EffTox                     | 22     | 19     | 10     | 12     | 29     | 8      |
| AIDE one_stage (Utility 2) | 10.8   | 43.5   | 17.2   | 6.3    | 1.5    | 20.7   |
| AIDE one_stage (Utility 3) | 12.1   | 46     | 15.5   | 4.7    | 0.9    | 20.8   |
| AIDE two_stage (Utility 2) | 11.4   | 31.7   | 11.8   | 4.6    | 2.4    | 38.1   |
| AIDE two_stage (Utility 3) | 12.4   | 32.5   | 10.7   | 4.6    | 1.5    | 38.3   |
| Mean patients treated      |        |        |        |        |        |        |
| BOIN12                     | 8.8    | 10.4   | 5.1    | 2.5    | 1.1    |        |
| U-BOIN                     | 7.7    | 9.2    | 6.5    | 3.6    | 2      |        |
| EffTox                     | 14.5   | 5.4    | 3.2    | 2.2    | 3.2    |        |
| AIDE one_stage (Utility 2) | 9.5    | 10.7   | 4.8    | 2.2    | 0.7    |        |
| AIDE one_stage (Utility 3) | 9.7    | 11.1   | 4.6    | 2      | 0.5    |        |
| AIDE two_stage (Utility 2) | 11.7   | 5.8    | 4.1    | 2.2    | 1.2    |        |
| AIDE two_stage (Utility 3) | 11.7   | 5.9    | 4      | 2.2    | 1.2    |        |
| Scenario 11                |        |        |        |        |        |        |
| DLT rate                   | 0.22   | 0.32   | 0.44   | 0.59   | 0.67   |        |
| Efficacy rate              | 0.2    | 0.2    | 0.2    | 0.2    | 0.2    |        |
| Utility 2                  | 0      | -0.1   | -0.2   | -0.4   | -0.5   |        |
| Utility 3                  | 43.2   | 39.2   | 34.4   | 28.4   | 25.2   |        |
| OBD selection (%)          |        |        |        |        |        |        |
| BOIN12                     | 64.7   | 23.1   | 4.4    | 0.2    | 0.1    | 7.5    |
| U-BOIN                     | 49.1   | 19.1   | 13.5   | 6.8    | 8      | 3.5    |
| EffTox                     | 69     | 5      | 3      | 2      | 1      | 20     |
| AIDE one_stage (Utility 2) | 66     | 23.5   | 5.3    | 0.5    | 0      | 4.7    |
| AIDE one_stage (Utility 3) | 69.7   | 20.4   | 4      | 0.6    | 0      | 5.3    |
| AIDE two_stage (Utility 2) | 65.9   | 22     | 4.6    | 0.5    | 0      | 7      |
| AIDE two_stage (Utility 3) | 69.5   | 20.6   | 2.4    | 0.7    | 0      | 6.8    |
| Mean patients treated      |        |        |        |        |        |        |
| BOIN12                     | 14.7   | 8      | 3.9    | 1.7    | 0.5    |        |
| U-BOIN                     | 11.9   | 6.6    | 4.7    | 3.2    | 2.8    |        |
| EffTox                     | 21.8   | 2.2    | 1.5    | 0.8    | 0.4    |        |
| AIDE one_stage (Utility 2) | 16.5   | 7.9    | 3.4    | 1.4    | 0.2    |        |
| AIDE one_stage (Utility 3) | 17.1   | 7.6    | 3.2    | 1.3    | 0.2    |        |
| AIDE two_stage (Utility 2) | 18.4   | 5.8    | 3.5    | 1.1    | 0.2    |        |
| AIDE two_stage (Utility 3) | 18.4   | 5.8    | 3.5    | 1.1    | 0.2    |        |
| Scenario 12                |        |        |        |        |        |        |
| DLT rate                   | 0.06   | 0.09   | 0.18   | 0.29   | 0.44   |        |
| Efficacy rate              | 0.05   | 0.1    | 0.12   | 0.4    | 0.6    |        |
| Utility 2                  | 0      | 0      | -0.1   | 0.1    | 0.2    |        |
| Utility 3                  | 40.6   | 42.4   | 40     | 52.4   | 58.4   |        |
| OBD selection (%)          |        |        |        |        |        |        |
| BOIN12                     | 20     | 26.2   | 13.1   | 31     | 9.5    | 0.2    |
| U-BOIN                     | 6.1    | 9.6    | 6.3    | 38.7   | 39.2   | 0.1    |
| EffTox                     | 10     | 9      | 24     | 18     | 36     | 4      |
| AIDE one_stage (Utility 2) | 11.7   | 11.2   | 8.4    | 44.2   | 24.1   | 0.4    |
| AIDE one_stage (Utility 3) | 16.6   | 14.3   | 6.7    | 41     | 21.2   | 0.2    |
| AIDE two_stage (Utility 2) | 14.2   | 12.3   | 10     | 36     | 24.2   | 3.3    |
| AIDE two_stage (Utility 3) | 21.2   | 12.2   | 6.8    | 34.4   | 22.1   | 3.3    |
| Mean patients treated      |        |        |        |        |        |        |
| BOIN12                     | 5.4    | 6.5    | 5.9    | 7.9    | 4.2    |        |
| U-BOIN                     | 5.2    | 6.3    | 6.4    | 7.4    | 4.9    |        |
| EffTox                     | 9.6    | 4.1    | 6.1    | 3.9    | 5.8    |        |
| AIDE one_stage (Utility 2) | 4.6    | 5.1    | 5.4    | 9.4    | 5.4    |        |
| AIDE one_stage (Utility 3) | 4.9    | 5.1    | 5.3    | 9.6    | 5.1    |        |
| AIDE two_stage (Utility 2) | 5      | 4.5    | 5.3    | 7.3    | 7.4    |        |
| AIDE two_stage (Utility 3) | 5      | 4.6    | 5.2    | 7.3    | 7.3    |        |



# Phase I/II Operating Characteristics

Method comparison - N = 30 (AIDE is an alpha-based model)

|                            | Dose 1 | Dose 2 | Dose 3 | Dose 4 | Dose 5 | Stop % |
|----------------------------|--------|--------|--------|--------|--------|--------|
| Scenario 13                |        |        |        |        |        |        |
| DLT rate                   | 0.21   | 0.29   | 0.3    | 0.39   | 0.46   |        |
| Efficacy rate              | 0.05   | 0.3    | 0.32   | 0.35   | 0.4    |        |
| Utility 2                  | -0.2   | 0      | 0      | 0      | -0.1   |        |
| Utility 3                  | 34.6   | 46.4   | 47.2   | 45.4   | 45.6   |        |
| OBD selection (%)          |        |        |        |        |        |        |
| BOIN12                     | 18.3   | 41.2   | 17.5   | 5      | 0.9    | 17.1   |
| U-BOIN                     | 4.3    | 21.5   | 27.2   | 21.4   | 21.3   | 4.3    |
| EffTox                     | 27     | 17     | 11     | 11     | 24     | 10     |
| AIDE one_stage (Utility 2) | 12.2   | 44.8   | 17.1   | 6.7    | 1.7    | 17.5   |
| AIDE one_stage (Utility 3) | 14.9   | 43.9   | 17     | 5.3    | 1.3    | 17.6   |
| AIDE two_stage (Utility 2) | 13.8   | 32.8   | 13.4   | 4.6    | 2      | 33.4   |
| AIDE two_stage (Utility 3) | 15.8   | 32.7   | 12.6   | 4.4    | 1.6    | 32.9   |
| Mean patients treated      |        |        |        |        |        |        |
| BOIN12                     | 9.5    | 9.1    | 5.3    | 3      | 1.3    |        |
| U-BOIN                     | 7.9    | 7.1    | 6.5    | 4.5    | 3      |        |
| EffTox                     | 14.5   | 5.4    | 3.2    | 2      | 2.8    |        |
| AIDE one_stage (Utility 2) | 9.5    | 9.9    | 5      | 3      | 0.9    |        |
| AIDE one_stage (Utility 3) | 9.8    | 10.2   | 4.9    | 2.7    | 0.8    |        |
| AIDE two_stage (Utility 2) | 11.4   | 6.1    | 4.5    | 2.5    | 1.2    |        |
| AIDE two_stage (Utility 3) | 11.4   | 6.1    | 4.5    | 2.4    | 1.2    |        |
| Scenario 14                |        |        |        |        |        |        |
| DLT rate                   | 0.09   | 0.15   | 0.23   | 0.28   | 0.39   |        |
| Efficacy rate              | 0.05   | 0.08   | 0.12   | 0.3    | 0.4    |        |
| Utility 2                  | 0      | -0.1   | -0.1   | 0      | 0      |        |
| Utility 3                  | 39.4   | 38.8   | 38     | 46.8   | 48.4   |        |
| OBD selection (%)          |        |        |        |        |        |        |
| BOIN12                     | 28.7   | 22.2   | 13.5   | 24.9   | 9      | 1.7    |
| U-BOIN                     | 11.9   | 9.5    | 9.2    | 33.9   | 35.1   | 0.4    |
| EffTox                     | 9      | 9      | 21     | 15     | 36     | 11     |
| AIDE one_stage (Utility 2) | 18.6   | 17.6   | 13.7   | 34.4   | 14.9   | 0.8    |
| AIDE one_stage (Utility 3) | 26.8   | 18.9   | 11.6   | 30.2   | 11.6   | 0.9    |
| AIDE two_stage (Utility 2) | 19.6   | 14.5   | 12.8   | 27.8   | 18     | 7.3    |
| AIDE two_stage (Utility 3) | 30.5   | 15.1   | 9      | 25.3   | 12.7   | 7.4    |
| Mean patients treated      |        |        |        |        |        |        |
| BOIN12                     | 6.6    | 6.8    | 5.8    | 6.6    | 4.1    |        |
| U-BOIN                     | 6.5    | 6.6    | 5.9    | 6.3    | 4.8    |        |
| EffTox                     | 9.4    | 3.9    | 5.8    | 3.6    | 5.8    |        |
| AIDE one_stage (Utility 2) | 6.2    | 6.1    | 5.6    | 7.7    | 4.3    |        |
| AIDE one_stage (Utility 3) | 6.8    | 6      | 5.4    | 7.7    | 4      |        |
| AIDE two_stage (Utility 2) | 6.7    | 5.2    | 5.8    | 6.1    | 5.3    |        |
| AIDE two_stage (Utility 3) | 6.7    | 5.2    | 5.8    | 6.1    | 5.2    |        |
| Scenario 15                |        |        |        |        |        |        |
| DLT rate                   | 0.13   | 0.2    | 0.32   | 0.39   | 0.46   |        |
| Efficacy rate              | 0.1    | 0.1    | 0.1    | 0.1    | 0.1    |        |
| Utility 2                  | 0      | -0.1   | -0.2   | -0.3   | -0.4   |        |
| Utility 3                  | 40.8   | 38     | 33.2   | 30.4   | 27.6   |        |
| OBD selection (%)          |        |        |        |        |        |        |
| BOIN12                     | 49.4   | 30.8   | 12.2   | 3.4    | 1.3    | 2.9    |
| U-BOIN                     | 41.1   | 23.2   | 11.8   | 11.2   | 12.2   | 0.5    |
| EffTox                     | 26     | 11     | 10     | 9      | 7      | 37     |
| AIDE one_stage (Utility 2) | 47.7   | 30.2   | 13.8   | 6.1    | 0.6    | 1.6    |
| AIDE one_stage (Utility 3) | 61.3   | 26.2   | 8      | 2.5    | 0.4    | 1.6    |
| AIDE two_stage (Utility 2) | 52.4   | 24.9   | 10.9   | 3.6    | 1.2    | 7      |
| AIDE two_stage (Utility 3) | 63.8   | 20.5   | 5.7    | 2.4    | 0.4    | 7.2    |
| Mean patients treated      |        |        |        |        |        |        |
| BOIN12                     | 10.3   | 8.4    | 5.5    | 3.6    | 2      |        |
| U-BOIN                     | 9      | 7.6    | 5.6    | 4.3    | 3.5    |        |
| EffTox                     | 14.1   | 3.2    | 3.7    | 2.1    | 1.8    |        |
| AIDE one_stage (Utility 2) | 10.1   | 8.2    | 5.7    | 4.2    | 1.6    |        |
| AIDE one_stage (Utility 3) | 11.1   | 7.9    | 5.4    | 4      | 1.5    |        |
| AIDE two_stage (Utility 2) | 11.6   | 6.6    | 5.9    | 3.3    | 1.7    |        |
| AIDE two_stage (Utility 3) | 11.7   | 6.6    | 5.8    | 3.3    | 1.7    |        |

# Phase I/II Operating Characteristics

Method comparison - N = 30 (AIDE is an alpha-based model)

|                            | Dose 1 | Dose 2 | Dose 3 | Dose 4 | Dose 5 | Stop % |
|----------------------------|--------|--------|--------|--------|--------|--------|
| Scenario 16                |        |        |        |        |        |        |
| DLT rate                   | 0.02   | 0.06   | 0.09   | 0.14   | 0.26   |        |
| Efficacy rate              | 0.2    | 0.2    | 0.2    | 0.2    | 0.2    |        |
| Utility 2                  | 0.2    | 0.1    | 0.1    | 0.1    | -0.1   |        |
| Utility 3                  | 51.2   | 49.6   | 48.4   | 46.4   | 41.6   |        |
| OBD selection (%)          |        |        |        |        |        |        |
| BOIN12                     | 36.8   | 26.2   | 20.4   | 11.7   | 4.9    | 0      |
| U-BOIN                     | 30.4   | 21.6   | 22.9   | 17.1   | 8      | 0      |
| EffTox                     | 66     | 5      | 8      | 7      | 9      | 5      |
| AIDE one_stage (Utility 2) | 41.3   | 25     | 13.3   | 13.7   | 6.7    | 0      |
| AIDE one_stage (Utility 3) | 46.9   | 24.5   | 13.3   | 11.2   | 4.1    | 0      |
| AIDE two_stage (Utility 2) | 31.4   | 23.2   | 16.2   | 16.6   | 12.5   | 0.1    |
| AIDE two_stage (Utility 3) | 38.2   | 25.7   | 15.6   | 12.9   | 7.5    | 0.1    |
| Mean patients treated      |        |        |        |        |        |        |
| BOIN12                     | 5      | 5.7    | 6.5    | 7.4    | 5.4    |        |
| U-BOIN                     | 6.1    | 6      | 6.2    | 6.4    | 5.6    |        |
| EffTox                     | 20.9   | 2.3    | 2.5    | 1.7    | 2.1    |        |
| AIDE one_stage (Utility 2) | 4.5    | 4.9    | 6      | 8.3    | 6.4    |        |
| AIDE one_stage (Utility 3) | 4.7    | 5.1    | 6.2    | 8.3    | 5.8    |        |
| AIDE two_stage (Utility 2) | 6.1    | 5.4    | 5.4    | 5.8    | 7.3    |        |
| AIDE two_stage (Utility 3) | 6      | 5.4    | 5.5    | 5.8    | 7.3    |        |
| Scenario 17                |        |        |        |        |        |        |
| DLT rate                   | 0.11   | 0.23   | 0.39   | 0.58   | 0.76   |        |
| Efficacy rate              | 0.3    | 0.5    | 0.5    | 0.5    | 0.5    |        |
| Utility 2                  | 0.2    | 0.3    | 0.1    | -0.1   | -0.3   |        |
| Utility 3                  | 53.6   | 60.8   | 54.4   | 46.8   | 39.6   |        |
| OBD selection (%)          |        |        |        |        |        |        |
| BOIN12                     | 33.7   | 55.7   | 9.2    | 0.7    | 0      | 0.7    |
| U-BOIN                     | 28.7   | 50.6   | 13.6   | 4.4    | 2.4    | 0.3    |
| EffTox                     | 91     | 5      | 3      | 1      | 0      | 1      |
| AIDE one_stage (Utility 2) | 20.8   | 58.9   | 17.3   | 2.8    | 0.1    | 0.1    |
| AIDE one_stage (Utility 3) | 24.4   | 60     | 13.7   | 1.8    | 0      | 0.1    |
| AIDE two_stage (Utility 2) | 30.3   | 49.6   | 17.9   | 1.9    | 0      | 0.3    |
| AIDE two_stage (Utility 3) | 34     | 49.1   | 14.8   | 1.8    | 0      | 0.3    |
| Mean patients treated      |        |        |        |        |        |        |
| BOIN12                     | 10     | 13.6   | 4.7    | 1.3    | 0.2    |        |
| U-BOIN                     | 10.8   | 12.1   | 5.1    | 1.5    | 0.4    |        |
| EffTox                     | 26.6   | 1.8    | 0.9    | 0.4    | 0.1    |        |
| AIDE one_stage (Utility 2) | 9.1    | 14.1   | 5.3    | 1.3    | 0.2    |        |
| AIDE one_stage (Utility 3) | 9.8    | 14.4   | 4.7    | 1      | 0.1    |        |
| AIDE two_stage (Utility 2) | 10.2   | 9.7    | 7.3    | 2.4    | 0.4    |        |
| AIDE two_stage (Utility 3) | 10.3   | 9.7    | 7.2    | 2.3    | 0.5    |        |
| Scenario 18                |        |        |        |        |        |        |
| DLT rate                   | 0.02   | 0.04   | 0.1    | 0.2    | 0.35   |        |
| Efficacy rate              | 0.05   | 0.1    | 0.15   | 0.2    | 0.5    |        |
| Utility 2                  | 0      | 0.1    | 0.1    | 0      | 0.1    |        |
| Utility 3                  | 42.2   | 44.4   | 45     | 44     | 56     |        |
| OBD selection (%)          |        |        |        |        |        |        |
| BOIN12                     | 15.8   | 19.6   | 21     | 15.5   | 28.1   | 0      |
| U-BOIN                     | 7.9    | 10.8   | 18.2   | 15.9   | 47.2   | 0      |
| EffTox                     | 6      | 7      | 21     | 18     | 42     | 6      |
| AIDE one_stage (Utility 2) | 11.6   | 11.3   | 7.5    | 14.5   | 55.1   | 0      |
| AIDE one_stage (Utility 3) | 17.1   | 13.5   | 9.1    | 12.3   | 48     | 0      |
| AIDE two_stage (Utility 2) | 9.8    | 7.5    | 9.2    | 14.2   | 58.5   | 0.8    |
| AIDE two_stage (Utility 3) | 16.1   | 10     | 9.6    | 11.8   | 51.7   | 0.8    |
| Mean patients treated      |        |        |        |        |        |        |
| BOIN12                     | 4.2    | 5.2    | 6      | 6.7    | 7.9    |        |
| U-BOIN                     | 4.2    | 5.4    | 6.8    | 7.1    | 6.8    |        |
| EffTox                     | 8.5    | 3.9    | 6.2    | 4.1    | 6.5    |        |
| AIDE one_stage (Utility 2) | 3.5    | 4      | 5.1    | 6.8    | 10.6   |        |
| AIDE one_stage (Utility 3) | 3.6    | 4.1    | 5.2    | 6.9    | 10.2   |        |
| AIDE two_stage (Utility 2) | 3.8    | 3.8    | 4.6    | 5.5    | 12.2   |        |
| AIDE two_stage (Utility 3) | 3.8    | 3.9    | 4.6    | 5.5    | 12.2   |        |

# Phase I/II Operating Characteristics

Method comparison - N = 30 (AIDE is an alpha-based model)

|                            | Dose 1 | Dose 2 | Dose 3 | Dose 4 | Dose 5 | Stop % |
|----------------------------|--------|--------|--------|--------|--------|--------|
| Scenario 19                |        |        |        |        |        |        |
| DLT rate                   | 0.11   | 0.26   | 0.39   | 0.6    | 0.77   |        |
| Efficacy rate              | 0.08   | 0.3    | 0.32   | 0.35   | 0.4    |        |
| Utility 2                  | 0      | 0      | -0.1   | -0.2   | -0.4   |        |
| Utility 3                  | 40.4   | 47.6   | 43.6   | 37     | 33.2   |        |
| OBD selection (%)          |        |        |        |        |        |        |
| BOIN12                     | 26.7   | 57     | 11.2   | 0.7    | 0      | 4.4    |
| U-BOIN                     | 16.5   | 39.9   | 22.5   | 11.5   | 9.3    | 0.3    |
| EffTox                     | 45     | 28     | 9      | 6      | 3      | 10     |
| AIDE one_stage (Utility 2) | 14.5   | 59     | 20.7   | 3.1    | 0.1    | 2.6    |
| AIDE one_stage (Utility 3) | 17.7   | 61.1   | 16     | 2.5    | 0.1    | 2.6    |
| AIDE two_stage (Utility 2) | 20.6   | 51.9   | 17.5   | 1.6    | 0      | 8.4    |
| AIDE two_stage (Utility 3) | 23.3   | 52.2   | 15.1   | 1.2    | 0      | 8.2    |
| Mean patients treated      |        |        |        |        |        |        |
| BOIN12                     | 10     | 12.1   | 5.4    | 1.8    | 0.4    |        |
| U-BOIN                     | 7.8    | 10.3   | 6.3    | 3.4    | 2.1    |        |
| EffTox                     | 17.6   | 5.9    | 3.2    | 1.3    | 0.6    |        |
| AIDE one_stage (Utility 2) | 8      | 12.6   | 6.6    | 2.3    | 0.4    |        |
| AIDE one_stage (Utility 3) | 8.4    | 12.8   | 6.2    | 2.1    | 0.3    |        |
| AIDE two_stage (Utility 2) | 9.2    | 10.2   | 7      | 2.2    | 0.4    |        |
| AIDE two_stage (Utility 3) | 9.1    | 10.2   | 7.1    | 2.3    | 0.4    |        |
| Scenario 20                |        |        |        |        |        |        |
| DLT rate                   | 0.07   | 0.14   | 0.31   | 0.41   | 0.52   |        |
| Efficacy rate              | 0.05   | 0.25   | 0.25   | 0.25   | 0.25   |        |
| Utility 2                  | 0      | 0.1    | -0.1   | -0.2   | -0.3   |        |
| Utility 3                  | 40.2   | 49.4   | 42.6   | 38.6   | 34.2   |        |
| OBD selection (%)          |        |        |        |        |        |        |
| BOIN12                     | 16.7   | 58     | 18.5   | 5.2    | 0.7    | 0.9    |
| U-BOIN                     | 10     | 45     | 16.8   | 14     | 14     | 0.2    |
| EffTox                     | 24     | 30     | 20     | 7      | 10     | 9      |
| AIDE one_stage (Utility 2) | 9.8    | 48.5   | 27.5   | 12.1   | 1.8    | 0.3    |
| AIDE one_stage (Utility 3) | 14     | 50.4   | 23.6   | 10.4   | 1.3    | 0.3    |
| AIDE two_stage (Utility 2) | 12.3   | 44.8   | 27.7   | 9.7    | 1.2    | 4.3    |
| AIDE two_stage (Utility 3) | 18.1   | 45.8   | 23.4   | 7.8    | 0.6    | 4.3    |
| Mean patients treated      |        |        |        |        |        |        |
| BOIN12                     | 6.7    | 10.7   | 6.8    | 4.1    | 1.7    |        |
| U-BOIN                     | 6.1    | 10.8   | 6.8    | 4      | 2.3    |        |
| EffTox                     | 13.5   | 6.6    | 5.1    | 2      | 1.7    |        |
| AIDE one_stage (Utility 2) | 4.8    | 10     | 8.2    | 5.2    | 1.9    |        |
| AIDE one_stage (Utility 3) | 5.3    | 10.4   | 7.8    | 4.8    | 1.6    |        |
| AIDE two_stage (Utility 2) | 5.6    | 8.7    | 8.3    | 4.6    | 2.2    |        |
| AIDE two_stage (Utility 3) | 5.7    | 8.5    | 8.3    | 4.6    | 2.3    |        |
| Scenario 21                |        |        |        |        |        |        |
| DLT rate                   | 0.37   | 0.56   | 0.69   | 0.83   | 0.88   |        |
| Efficacy rate              | 0.5    | 0.5    | 0.5    | 0.5    | 0.5    |        |
| Utility 2                  | 0.1    | -0.1   | -0.2   | -0.3   | -0.4   |        |
| Utility 3                  | 55.2   | 47.6   | 42.4   | 36.8   | 34.8   |        |
| OBD selection (%)          |        |        |        |        |        |        |
| BOIN12                     | 58.5   | 0.9    | 0.1    | 0      | 0      | 40.5   |
| U-BOIN                     | 49.7   | 8.4    | 7.2    | 4.9    | 5.8    | 24     |
| EffTox                     | 72     | 0      | 0      | 0      | 0      | 28     |
| AIDE one_stage (Utility 2) | 57.1   | 1.8    | 0      | 0      | 0      | 41.1   |
| AIDE one_stage (Utility 3) | 57.9   | 1.5    | 0      | 0      | 0      | 40.6   |
| AIDE two_stage (Utility 2) | 59.9   | 1.6    | 0      | 0      | 0      | 38.5   |
| AIDE two_stage (Utility 3) | 60.5   | 1.4    | 0      | 0      | 0      | 38.1   |
| Mean patients treated      |        |        |        |        |        |        |
| BOIN12                     | 19.9   | 2.6    | 0.4    | 0      | 0      |        |
| U-BOIN                     | 17     | 4      | 1.6    | 1      | 0.8    |        |
| EffTox                     | 25     | 0.3    | 0.1    | 0      | 0      |        |
| AIDE one_stage (Utility 2) | 20.2   | 2.5    | 0.2    | 0      | 0      |        |
| AIDE one_stage (Utility 3) | 20.4   | 2.5    | 0.2    | 0      | 0      |        |
| AIDE two_stage (Utility 2) | 20.9   | 2.1    | 0.6    | 0      | 0      |        |
| AIDE two_stage (Utility 3) | 20.9   | 2      | 0.6    | 0      | 0      |        |

# Phase I/II Operating Characteristics

Method comparison - N = 30 (AIDE is an alpha-based model)

|                            | Dose 1 | Dose 2 | Dose 3 | Dose 4 | Dose 5 | Stop % |
|----------------------------|--------|--------|--------|--------|--------|--------|
| Scenario 22                |        |        |        |        |        |        |
| DLT rate                   | 0.07   | 0.27   | 0.5    | 0.56   | 0.78   |        |
| Efficacy rate              | 0.08   | 0.25   | 0.25   | 0.5    | 0.5    |        |
| Utility 2                  | 0      | 0      | -0.2   | -0.1   | -0.3   |        |
| Utility 3                  | 42     | 44.2   | 35     | 47.6   | 38.8   |        |
| OBD selection (%)          |        |        |        |        |        |        |
| BOIN12                     | 40.8   | 50.2   | 3.8    | 0.6    | 0      | 4.6    |
| U-BOIN                     | 25.2   | 30.3   | 10     | 23.2   | 11.1   | 0.2    |
| EffTox                     | 39     | 29     | 10     | 4      | 2      | 16     |
| AIDE one_stage (Utility 2) | 18.4   | 62.5   | 14.1   | 3.7    | 0      | 1.3    |
| AIDE one_stage (Utility 3) | 23     | 60.5   | 11.4   | 3.5    | 0      | 1.6    |
| AIDE two_stage (Utility 2) | 26.3   | 55.1   | 12.4   | 1.9    | 0      | 4.3    |
| AIDE two_stage (Utility 3) | 29.5   | 54.7   | 10     | 1.5    | 0      | 4.3    |
| Mean patients treated      |        |        |        |        |        |        |
| BOIN12                     | 11.6   | 11.4   | 4.4    | 2      | 0.3    |        |
| U-BOIN                     | 8.3    | 9.7    | 5      | 4.6    | 2.4    |        |
| EffTox                     | 17     | 5.9    | 3.3    | 1.2    | 0.5    |        |
| AIDE one_stage (Utility 2) | 8.5    | 13.3   | 5.7    | 2.2    | 0.2    |        |
| AIDE one_stage (Utility 3) | 9.2    | 13.1   | 5.4    | 2.1    | 0.2    |        |
| AIDE two_stage (Utility 2) | 8.7    | 12.1   | 6.5    | 1.7    | 0.3    |        |
| AIDE two_stage (Utility 3) | 8.7    | 12.1   | 6.6    | 1.7    | 0.3    |        |
| Scenario 23                |        |        |        |        |        |        |
| DLT rate                   | 0.32   | 0.55   | 0.8    | 0.94   | 0.97   |        |
| Efficacy rate              | 0.05   | 0.05   | 0.05   | 0.05   | 0.05   |        |
| Utility 2                  | -0.3   | -0.5   | -0.8   | -0.9   | -0.9   |        |
| Utility 3                  | 30.2   | 21     | 11     | 5.4    | 4.2    |        |
| OBD selection (%)          |        |        |        |        |        |        |
| BOIN12                     | 24.5   | 3      | 0      | 0      | 0      | 72.5   |
| U-BOIN                     | 47.6   | 18.1   | 5.9    | 1.9    | 0.5    | 26     |
| EffTox                     | 5      | 4      | 4      | 1      | 0      | 86     |
| AIDE one_stage (Utility 2) | 21     | 4.2    | 0      | 0      | 0      | 74.8   |
| AIDE one_stage (Utility 3) | 20.9   | 3.9    | 0.1    | 0      | 0      | 75.1   |
| AIDE two_stage (Utility 2) | 19.6   | 5.8    | 0.2    | 0      | 0      | 74.4   |
| AIDE two_stage (Utility 3) | 20.2   | 5.7    | 0      | 0      | 0      | 74.1   |
| Mean patients treated      |        |        |        |        |        |        |
| BOIN12                     | 16.1   | 4.7    | 1.1    | 0.1    | 0      |        |
| U-BOIN                     | 13.3   | 4.7    | 2.9    | 2.6    | 2.6    |        |
| EffTox                     | 8      | 3.1    | 1.7    | 0.3    | 0      |        |
| AIDE one_stage (Utility 2) | 15.9   | 4.8    | 0.7    | 0.1    | 0      |        |
| AIDE one_stage (Utility 3) | 15.9   | 4.7    | 0.7    | 0.1    | 0      |        |
| AIDE two_stage (Utility 2) | 17.2   | 2.7    | 0.6    | 0      | 0      |        |
| AIDE two_stage (Utility 3) | 17.2   | 2.7    | 0.6    | 0      | 0      |        |
| Scenario 24                |        |        |        |        |        |        |
| DLT rate                   | 0.01   | 0.02   | 0.05   | 0.1    | 0.25   |        |
| Efficacy rate              | 0.05   | 0.08   | 0.2    | 0.5    | 0.5    |        |
| Utility 2                  | 0      | 0.1    | 0.1    | 0.4    | 0.2    |        |
| Utility 3                  | 42.6   | 44     | 50     | 66     | 60     |        |
| OBD selection (%)          |        |        |        |        |        |        |
| BOIN12                     | 2.6    | 2.1    | 6.9    | 62.8   | 25.6   | 0      |
| U-BOIN                     | 1.2    | 1.7    | 7.3    | 60.4   | 29.4   | 0      |
| EffTox                     | 5      | 5      | 10     | 18     | 63     | 1      |
| AIDE one_stage (Utility 2) | 2.3    | 1.4    | 3.1    | 56.3   | 36.9   | 0      |
| AIDE one_stage (Utility 3) | 3.9    | 3      | 4.8    | 58.6   | 29.7   | 0      |
| AIDE two_stage (Utility 2) | 0.9    | 1.9    | 4.8    | 43.8   | 48.5   | 0.1    |
| AIDE two_stage (Utility 3) | 2.4    | 2.4    | 6.5    | 47.4   | 41.2   | 0.1    |
| Mean patients treated      |        |        |        |        |        |        |
| BOIN12                     | 3.2    | 3.4    | 4.5    | 12.5   | 6.3    |        |
| U-BOIN                     | 3.4    | 3.9    | 5      | 9.7    | 8.3    |        |
| EffTox                     | 8.1    | 3.3    | 4.6    | 4      | 9.9    |        |
| AIDE one_stage (Utility 2) | 3.1    | 3.1    | 4      | 12.5   | 7.2    |        |
| AIDE one_stage (Utility 3) | 3.1    | 3.2    | 4.1    | 13.1   | 6.5    |        |
| AIDE two_stage (Utility 2) | 3.3    | 3.3    | 3.8    | 8.2    | 11.4   |        |
| AIDE two_stage (Utility 3) | 3.3    | 3.3    | 3.8    | 8.2    | 11.4   |        |



# Phase I/II Operating Characteristics

Method comparison - N = 30 (AIDE is an alpha-based model)

|                            | Dose 1 | Dose 2 | Dose 3 | Dose 4 | Dose 5 | Stop % |
|----------------------------|--------|--------|--------|--------|--------|--------|
| Scenario 25                |        |        |        |        |        |        |
| DLT rate                   | 0.09   | 0.18   | 0.37   | 0.63   | 0.77   |        |
| Efficacy rate              | 0.2    | 0.25   | 0.4    | 0.45   | 0.5    |        |
| Utility 2                  | 0.1    | 0.1    | 0      | -0.2   | -0.3   |        |
| Utility 3                  | 48.4   | 47.8   | 49.2   | 41.8   | 39.2   |        |
| OBD selection (%)          |        |        |        |        |        |        |
| BOIN12                     | 43.2   | 36     | 20.1   | 0.6    | 0      | 0.1    |
| U-BOIN                     | 30     | 25.3   | 26.2   | 9.9    | 8.5    | 0.1    |
| EffTox                     | 77     | 12     | 5      | 3      | 1      | 3      |
| AIDE one_stage (Utility 2) | 28.1   | 28.6   | 37     | 6.1    | 0.1    | 0.1    |
| AIDE one_stage (Utility 3) | 32.8   | 30.6   | 32     | 4.5    | 0      | 0.1    |
| AIDE two_stage (Utility 2) | 34     | 28.3   | 32.3   | 4.7    | 0      | 0.7    |
| AIDE two_stage (Utility 3) | 38.7   | 28.9   | 27.9   | 3.8    | 0      | 0.7    |
| Mean patients treated      |        |        |        |        |        |        |
| BOIN12                     | 10.2   | 9.8    | 7.5    | 2.2    | 0.3    |        |
| U-BOIN                     | 9.7    | 9      | 7.2    | 2.9    | 1.2    |        |
| EffTox                     | 22.9   | 3.3    | 2      | 1      | 0.3    |        |
| AIDE one_stage (Utility 2) | 8.9    | 9      | 8.8    | 2.8    | 0.4    |        |
| AIDE one_stage (Utility 3) | 9.6    | 9.2    | 8.3    | 2.5    | 0.3    |        |
| AIDE two_stage (Utility 2) | 9.3    | 7.8    | 9.4    | 2.8    | 0.5    |        |
| AIDE two_stage (Utility 3) | 9.4    | 7.8    | 9.4    | 2.8    | 0.5    |        |
| Scenario 26                |        |        |        |        |        |        |
| DLT rate                   | 0.15   | 0.22   | 0.3    | 0.62   | 0.8    |        |
| Efficacy rate              | 0.1    | 0.2    | 0.35   | 0.45   | 0.55   |        |
| Utility 2                  | -0.1   | 0      | 0.1    | -0.2   | -0.2   |        |
| Utility 3                  | 40     | 43.2   | 49     | 42.2   | 41     |        |
| OBD selection (%)          |        |        |        |        |        |        |
| BOIN12                     | 31.1   | 32.3   | 32.3   | 0.9    | 0      | 3.4    |
| U-BOIN                     | 15.8   | 22.3   | 36.7   | 12.9   | 11.2   | 1.1    |
| EffTox                     | 38     | 25     | 18     | 8      | 4      | 8      |
| AIDE one_stage (Utility 2) | 22.8   | 32.2   | 34.1   | 7.4    | 0.1    | 3.4    |
| AIDE one_stage (Utility 3) | 28.6   | 31.3   | 30.2   | 6.5    | 0.1    | 3.3    |
| AIDE two_stage (Utility 2) | 29     | 26.5   | 28.5   | 5.8    | 0.1    | 10.1   |
| AIDE two_stage (Utility 3) | 34.3   | 26.1   | 24.2   | 5.2    | 0      | 10.2   |
| Mean patients treated      |        |        |        |        |        |        |
| BOIN12                     | 9.4    | 8.9    | 8.1    | 2.9    | 0.4    |        |
| U-BOIN                     | 7.9    | 7.6    | 8      | 4.1    | 2.1    |        |
| EffTox                     | 17     | 4.9    | 3.8    | 2.1    | 1      |        |
| AIDE one_stage (Utility 2) | 8.8    | 8.9    | 8.4    | 3.2    | 0.4    |        |
| AIDE one_stage (Utility 3) | 9.4    | 8.8    | 8.1    | 2.9    | 0.4    |        |
| AIDE two_stage (Utility 2) | 11     | 6.5    | 8      | 2.8    | 0.5    |        |
| AIDE two_stage (Utility 3) | 11     | 6.5    | 7.9    | 2.9    | 0.5    |        |
| Scenario 27                |        |        |        |        |        |        |
| DLT rate                   | 0.11   | 0.17   | 0.3    | 0.5    | 0.62   |        |
| Efficacy rate              | 0.1    | 0.2    | 0.4    | 0.42   | 0.45   |        |
| Utility 2                  | 0      | 0      | 0.1    | -0.1   | -0.2   |        |
| Utility 3                  | 41.6   | 45.2   | 52     | 45.2   | 42.2   |        |
| OBD selection (%)          |        |        |        |        |        |        |
| BOIN12                     | 22.1   | 32.1   | 40.3   | 3.5    | 0.2    | 1.8    |
| U-BOIN                     | 9.9    | 20.2   | 39.6   | 15.7   | 14.4   | 0.2    |
| EffTox                     | 36     | 27     | 15     | 12     | 6      | 4      |
| AIDE one_stage (Utility 2) | 14.6   | 26.6   | 46.7   | 11     | 0.5    | 0.6    |
| AIDE one_stage (Utility 3) | 20.7   | 26.4   | 42.6   | 9.1    | 0.4    | 0.8    |
| AIDE two_stage (Utility 2) | 21.2   | 21.2   | 41.9   | 9.2    | 1      | 5.5    |
| AIDE two_stage (Utility 3) | 25.8   | 22.9   | 37.6   | 7.7    | 0.6    | 5.4    |
| Mean patients treated      |        |        |        |        |        |        |
| BOIN12                     | 7.5    | 9      | 9.1    | 3.3    | 0.8    |        |
| U-BOIN                     | 6.9    | 7.9    | 8.9    | 4.4    | 1.8    |        |
| EffTox                     | 16.6   | 5.1    | 3.7    | 2.2    | 1.7    |        |
| AIDE one_stage (Utility 2) | 6.6    | 8.2    | 10.2   | 4      | 0.9    |        |
| AIDE one_stage (Utility 3) | 7.2    | 8.1    | 10.1   | 3.8    | 0.7    |        |
| AIDE two_stage (Utility 2) | 8.3    | 6.3    | 9.4    | 4      | 1.4    |        |
| AIDE two_stage (Utility 3) | 8.2    | 6.4    | 9.4    | 4      | 1.4    |        |

# Phase I/II Operating Characteristics

Method comparison - N = 30 (AIDE is an alpha-based model)

|                            | Dose 1 | Dose 2 | Dose 3 | Dose 4 | Dose 5 | Stop % |
|----------------------------|--------|--------|--------|--------|--------|--------|
| Scenario 28                |        |        |        |        |        |        |
| DLT rate                   | 0.2    | 0.3    | 0.5    | 0.6    | 0.77   |        |
| Efficacy rate              | 0.05   | 0.3    | 0.32   | 0.35   | 0.4    |        |
| Utility 2                  | -0.1   | 0      | -0.2   | -0.2   | -0.4   |        |
| Utility 3                  | 35     | 46     | 39.2   | 37     | 33.2   |        |
| OBD selection (%)          |        |        |        |        |        |        |
| BOIN12                     | 25.1   | 50.6   | 4.4    | 0.5    | 0      | 19.4   |
| U-BOIN                     | 16.7   | 39.7   | 15.4   | 12.7   | 11.3   | 4.2    |
| EffTox                     | 36     | 29     | 8      | 5      | 4      | 19     |
| AIDE one_stage (Utility 2) | 18.7   | 52     | 9.4    | 0.5    | 0      | 19.4   |
| AIDE one_stage (Utility 3) | 20.6   | 51.6   | 7.8    | 0.3    | 0      | 19.7   |
| AIDE two_stage (Utility 2) | 21.6   | 37.6   | 6.8    | 0.3    | 0      | 33.7   |
| AIDE two_stage (Utility 3) | 21.8   | 38.8   | 5.7    | 0.3    | 0      | 33.4   |
| Mean patients treated      |        |        |        |        |        |        |
| BOIN12                     | 11.5   | 11.1   | 4      | 1.3    | 0.2    |        |
| U-BOIN                     | 8.4    | 9      | 5.5    | 3.5    | 2.5    |        |
| EffTox                     | 15.7   | 6.1    | 3.3    | 1.4    | 0.7    |        |
| AIDE one_stage (Utility 2) | 10.7   | 12.1   | 4.1    | 1.2    | 0.2    |        |
| AIDE one_stage (Utility 3) | 11     | 12     | 3.9    | 1.1    | 0.1    |        |
| AIDE two_stage (Utility 2) | 12     | 8.4    | 4.3    | 0.9    | 0.1    |        |
| AIDE two_stage (Utility 3) | 12.1   | 8.4    | 4.3    | 0.9    | 0.1    |        |
| Scenario 29                |        |        |        |        |        |        |
| DLT rate                   | 0.19   | 0.3    | 0.56   | 0.74   | 0.86   |        |
| Efficacy rate              | 0.4    | 0.4    | 0.4    | 0.4    | 0.4    |        |
| Utility 2                  | 0.2    | 0.1    | -0.2   | -0.3   | -0.5   |        |
| Utility 3                  | 56.4   | 52     | 41.6   | 34.4   | 29.6   |        |
| OBD selection (%)          |        |        |        |        |        |        |
| BOIN12                     | 71     | 24.8   | 1.2    | 0      | 0      | 3      |
| U-BOIN                     | 61.5   | 24.3   | 4.9    | 3.8    | 3.6    | 1.9    |
| EffTox                     | 96     | 1      | 0      | 0      | 0      | 3      |
| AIDE one_stage (Utility 2) | 66.1   | 27.1   | 4.2    | 0.2    | 0      | 2.4    |
| AIDE one_stage (Utility 3) | 69.6   | 25.2   | 2.8    | 0.1    | 0      | 2.3    |
| AIDE two_stage (Utility 2) | 69.7   | 23.3   | 4.1    | 0.3    | 0      | 2.6    |
| AIDE two_stage (Utility 3) | 72.4   | 21.5   | 3.2    | 0.3    | 0      | 2.6    |
| Mean patients treated      |        |        |        |        |        |        |
| BOIN12                     | 17.1   | 8.8    | 2.8    | 0.6    | 0      |        |
| U-BOIN                     | 16.2   | 8.8    | 3      | 1      | 0.6    |        |
| EffTox                     | 28.3   | 0.7    | 0.3    | 0.1    | 0      |        |
| AIDE one_stage (Utility 2) | 17.7   | 8.5    | 2.7    | 0.6    | 0      |        |
| AIDE one_stage (Utility 3) | 18.7   | 8.2    | 2.2    | 0.5    | 0      |        |
| AIDE two_stage (Utility 2) | 18.2   | 7.1    | 3.6    | 0.5    | 0      |        |
| AIDE two_stage (Utility 3) | 18.3   | 7.1    | 3.5    | 0.6    | 0      |        |
| Scenario 30                |        |        |        |        |        |        |
| DLT rate                   | 0.05   | 0.08   | 0.17   | 0.27   | 0.54   |        |
| Efficacy rate              | 0.12   | 0.15   | 0.2    | 0.4    | 0.5    |        |
| Utility 2                  | 0.1    | 0.1    | 0      | 0.1    | 0      |        |
| Utility 3                  | 45.2   | 45.8   | 45.2   | 53.2   | 48.4   |        |
| OBD selection (%)          |        |        |        |        |        |        |
| BOIN12                     | 22.4   | 21.9   | 19.1   | 33.7   | 2.9    | 0      |
| U-BOIN                     | 12.4   | 14.7   | 17.5   | 38.8   | 16.6   | 0      |
| EffTox                     | 39     | 14     | 20     | 14     | 11     | 3      |
| AIDE one_stage (Utility 2) | 17.2   | 11.9   | 11.2   | 45.3   | 14.4   | 0      |
| AIDE one_stage (Utility 3) | 27.9   | 14     | 9.8    | 38.3   | 10     | 0      |
| AIDE two_stage (Utility 2) | 21     | 13.2   | 15.7   | 37     | 12.3   | 0.8    |
| AIDE two_stage (Utility 3) | 27.8   | 14.5   | 13.8   | 32.9   | 10.2   | 0.8    |
| Mean patients treated      |        |        |        |        |        |        |
| BOIN12                     | 5.2    | 6.1    | 6.7    | 8.8    | 3.2    |        |
| U-BOIN                     | 5.4    | 6.6    | 7.3    | 7.4    | 3.7    |        |
| EffTox                     | 15.9   | 3.5    | 4.6    | 2.8    | 2.8    |        |
| AIDE one_stage (Utility 2) | 4.3    | 4.8    | 6      | 10.3   | 4.6    |        |
| AIDE one_stage (Utility 3) | 4.6    | 4.9    | 5.9    | 10.3   | 4.2    |        |
| AIDE two_stage (Utility 2) | 5.2    | 4.5    | 5.7    | 8.1    | 6.5    |        |
| AIDE two_stage (Utility 3) | 5.2    | 4.5    | 5.7    | 8.1    | 6.5    |        |

# Phase I/II Operating Characteristics

Method comparison - N = 30 (AIDE is an alpha-based model)

|                            | Dose 1 | Dose 2 | Dose 3 | Dose 4 | Dose 5 | Stop % |
|----------------------------|--------|--------|--------|--------|--------|--------|
| Scenario 31                |        |        |        |        |        |        |
| DLT rate                   | 0.04   | 0.06   | 0.1    | 0.2    | 0.6    |        |
| Efficacy rate              | 0.2    | 0.5    | 0.5    | 0.5    | 0.5    |        |
| Utility 2                  | 0.2    | 0.4    | 0.4    | 0.3    | -0.1   |        |
| Utility 3                  | 50.4   | 67.6   | 66     | 62     | 46     |        |
| OBD selection (%)          |        |        |        |        |        |        |
| BOIN12                     | 3.1    | 41.6   | 34.4   | 20.1   | 0.8    | 0      |
| U-BOIN                     | 4.6    | 39.7   | 34.4   | 20     | 1.3    | 0      |
| EffTox                     | 62     | 9      | 15     | 12     | 3      | 0      |
| AIDE one_stage (Utility 2) | 4.9    | 35.7   | 26.7   | 26.2   | 6.5    | 0      |
| AIDE one_stage (Utility 3) | 7.7    | 42.5   | 25.4   | 20.2   | 4.2    | 0      |
| AIDE two_stage (Utility 2) | 7.6    | 35     | 25.6   | 25.2   | 6.5    | 0.1    |
| AIDE two_stage (Utility 3) | 10.3   | 41.9   | 24.3   | 19.5   | 3.9    | 0.1    |
| Mean patients treated      |        |        |        |        |        |        |
| BOIN12                     | 3.8    | 12.3   | 8.1    | 4.6    | 1.1    |        |
| U-BOIN                     | 4.5    | 7.6    | 7.8    | 7.5    | 3      |        |
| EffTox                     | 20.7   | 2.9    | 3.1    | 2.3    | 1      |        |
| AIDE one_stage (Utility 2) | 3.8    | 11.1   | 7.2    | 5.8    | 2.1    |        |
| AIDE one_stage (Utility 3) | 3.9    | 12     | 7.1    | 5.3    | 1.6    |        |
| AIDE two_stage (Utility 2) | 4.1    | 5.7    | 6.4    | 7.7    | 6.1    |        |
| AIDE two_stage (Utility 3) | 4.1    | 5.7    | 6.4    | 7.6    | 6      |        |
| Scenario 32                |        |        |        |        |        |        |
| DLT rate                   | 0.02   | 0.05   | 0.11   | 0.3    | 0.36   |        |
| Efficacy rate              | 0.08   | 0.15   | 0.3    | 0.3    | 0.3    |        |
| Utility 2                  | 0.1    | 0.1    | 0.2    | 0      | -0.1   |        |
| Utility 3                  | 44     | 47     | 53.6   | 46     | 43.6   |        |
| OBD selection (%)          |        |        |        |        |        |        |
| BOIN12                     | 14.5   | 20.5   | 46.3   | 15     | 3.7    | 0      |
| U-BOIN                     | 6.5    | 16.9   | 45.1   | 17.9   | 13.6   | 0      |
| EffTox                     | 19     | 18     | 30     | 14     | 15     | 4      |
| AIDE one_stage (Utility 2) | 13.6   | 15.1   | 33.3   | 26.2   | 11.8   | 0      |
| AIDE one_stage (Utility 3) | 20     | 16.3   | 33.3   | 23.4   | 7      | 0      |
| AIDE two_stage (Utility 2) | 9.5    | 13.2   | 35     | 27.3   | 14.4   | 0.6    |
| AIDE two_stage (Utility 3) | 15.4   | 16.5   | 34     | 24.1   | 9.4    | 0.6    |
| Mean patients treated      |        |        |        |        |        |        |
| BOIN12                     | 4.3    | 5.9    | 9.5    | 6.7    | 3.6    |        |
| U-BOIN                     | 4.4    | 6      | 9.8    | 6.6    | 3.4    |        |
| EffTox                     | 12.1   | 4.7    | 5.9    | 3.6    | 3.2    |        |
| AIDE one_stage (Utility 2) | 3.5    | 4.3    | 8.5    | 8.5    | 5.3    |        |
| AIDE one_stage (Utility 3) | 3.6    | 4.5    | 8.9    | 8.4    | 4.6    |        |
| AIDE two_stage (Utility 2) | 4      | 4.3    | 7.6    | 7      | 7      |        |
| AIDE two_stage (Utility 3) | 4      | 4.3    | 7.6    | 7      | 7      |        |
| Scenario 33                |        |        |        |        |        |        |
| DLT rate                   | 0.01   | 0.02   | 0.09   | 0.32   | 0.38   |        |
| Efficacy rate              | 0.05   | 0.08   | 0.2    | 0.4    | 0.5    |        |
| Utility 2                  | 0      | 0.1    | 0.1    | 0.1    | 0.1    |        |
| Utility 3                  | 42.6   | 44     | 48.4   | 51.2   | 54.8   |        |
| OBD selection (%)          |        |        |        |        |        |        |
| BOIN12                     | 14.7   | 15.1   | 28.2   | 30.3   | 11.7   | 0      |
| U-BOIN                     | 6      | 7.9    | 27.8   | 28.5   | 29.8   | 0      |
| EffTox                     | 6      | 9      | 34     | 21     | 29     | 1      |
| AIDE one_stage (Utility 2) | 8.3    | 5.1    | 10.4   | 45.1   | 31.1   | 0      |
| AIDE one_stage (Utility 3) | 12.6   | 7.5    | 12.7   | 43     | 24.2   | 0      |
| AIDE two_stage (Utility 2) | 7.3    | 5.6    | 12.7   | 40     | 34.3   | 0.1    |
| AIDE two_stage (Utility 3) | 12.6   | 6.6    | 12.4   | 40.6   | 27.8   | 0      |
| Mean patients treated      |        |        |        |        |        |        |
| BOIN12                     | 4.5    | 5.6    | 7.3    | 8.2    | 4.4    |        |
| U-BOIN                     | 3.9    | 5      | 8.7    | 8      | 4.5    |        |
| EffTox                     | 8.9    | 4.1    | 7.1    | 4.5    | 5.3    |        |
| AIDE one_stage (Utility 2) | 3.3    | 3.6    | 5.9    | 10.3   | 6.8    |        |
| AIDE one_stage (Utility 3) | 3.5    | 3.7    | 6      | 10.5   | 6.3    |        |
| AIDE two_stage (Utility 2) | 3.5    | 3.6    | 5.2    | 8      | 9.7    |        |
| AIDE two_stage (Utility 3) | 3.5    | 3.7    | 5.2    | 8      | 9.7    |        |

# Phase I/II Operating Characteristics

Method comparison - N = 30 (AIDE is an alpha-based model)

|                            | Dose 1 | Dose 2 | Dose 3 | Dose 4 | Dose 5 | Stop % |
|----------------------------|--------|--------|--------|--------|--------|--------|
| Scenario 34                |        |        |        |        |        |        |
| DLT rate                   | 0.05   | 0.13   | 0.3    | 0.36   | 0.42   |        |
| Efficacy rate              | 0.05   | 0.1    | 0.3    | 0.32   | 0.35   |        |
| Utility 2                  | 0      | 0      | 0      | 0      | -0.1   |        |
| Utility 3                  | 41     | 40.8   | 46     | 44.8   | 44.2   |        |
| OBD selection (%)          |        |        |        |        |        |        |
| BOIN12                     | 28.2   | 22.3   | 35.3   | 10.8   | 2.8    | 0.6    |
| U-BOIN                     | 10.7   | 10.4   | 28.2   | 23     | 27.7   | 0      |
| EffTox                     | 12     | 23     | 27     | 11     | 16     | 10     |
| AIDE one_stage (Utility 2) | 14.4   | 12.8   | 44.8   | 22.3   | 5.5    | 0.2    |
| AIDE one_stage (Utility 3) | 17.9   | 15.2   | 43     | 20.4   | 3.3    | 0.2    |
| AIDE two_stage (Utility 2) | 13.4   | 12.2   | 47.8   | 18.4   | 6.2    | 2      |
| AIDE two_stage (Utility 3) | 19.7   | 13.9   | 44.9   | 16.4   | 3.2    | 1.9    |
| Mean patients treated      |        |        |        |        |        |        |
| BOIN12                     | 7.4    | 8      | 7.6    | 4.7    | 2.3    |        |
| U-BOIN                     | 5.8    | 7.4    | 8.1    | 5.3    | 3.5    |        |
| EffTox                     | 10.1   | 5.9    | 6.4    | 2.9    | 3.4    |        |
| AIDE one_stage (Utility 2) | 5.1    | 6.3    | 9.6    | 6.2    | 2.8    |        |
| AIDE one_stage (Utility 3) | 5.8    | 6.3    | 9.5    | 6      | 2.4    |        |
| AIDE two_stage (Utility 2) | 5.3    | 5.6    | 10     | 5.3    | 3.5    |        |
| AIDE two_stage (Utility 3) | 5.4    | 5.7    | 9.9    | 5.3    | 3.5    |        |
| Scenario 35                |        |        |        |        |        |        |
| DLT rate                   | 0.03   | 0.09   | 0.3    | 0.4    | 0.45   |        |
| Efficacy rate              | 0.2    | 0.45   | 0.45   | 0.45   | 0.45   |        |
| Utility 2                  | 0.2    | 0.4    | 0.1    | 0.1    | 0      |        |
| Utility 3                  | 50.8   | 63.4   | 55     | 51     | 49     |        |
| OBD selection (%)          |        |        |        |        |        |        |
| BOIN12                     | 10.5   | 62.5   | 21.7   | 4.1    | 1.2    | 0      |
| U-BOIN                     | 9.9    | 61.2   | 20.2   | 5.9    | 2.8    | 0      |
| EffTox                     | 66     | 21     | 8      | 4      | 2      | 0      |
| AIDE one_stage (Utility 2) | 10.2   | 40.3   | 31.5   | 14.3   | 3.7    | 0      |
| AIDE one_stage (Utility 3) | 13.8   | 46.6   | 27.7   | 9.4    | 2.5    | 0      |
| AIDE two_stage (Utility 2) | 12.8   | 39.8   | 31.2   | 11.9   | 4.2    | 0.1    |
| AIDE two_stage (Utility 3) | 16.4   | 44.1   | 26.4   | 10.4   | 2.6    | 0.1    |
| Mean patients treated      |        |        |        |        |        |        |
| BOIN12                     | 5.3    | 13.8   | 6.8    | 2.9    | 1.1    |        |
| U-BOIN                     | 5.5    | 12.8   | 7.8    | 3      | 1      |        |
| EffTox                     | 21.4   | 4.3    | 2.7    | 1.1    | 0.4    |        |
| AIDE one_stage (Utility 2) | 4.4    | 11.2   | 8.5    | 4.4    | 1.6    |        |
| AIDE one_stage (Utility 3) | 4.6    | 12.2   | 7.9    | 4      | 1.3    |        |
| AIDE two_stage (Utility 2) | 4.7    | 8.2    | 8.9    | 5      | 3.2    |        |
| AIDE two_stage (Utility 3) | 4.7    | 8.3    | 8.9    | 5      | 3.2    |        |



# Phase I/II Operating Characteristics

Method comparison - N = 30 (AIDE is an alpha-based model)

|                            | Dose 1 | Dose 2 | Dose 3 | Dose 4 | Dose 5 | Stop % |
|----------------------------|--------|--------|--------|--------|--------|--------|
| Scenario 36                |        |        |        |        |        |        |
| DLT rate                   | 0.13   | 0.29   | 0.35   | 0.4    | 0.43   |        |
| Efficacy rate              | 0.05   | 0.2    | 0.22   | 0.3    | 0.35   |        |
| Utility 2                  | -0.1   | -0.1   | -0.1   | -0.1   | -0.1   |        |
| Utility 3                  | 37.8   | 40.4   | 39.2   | 42     | 43.8   |        |
| OBD selection (%)          |        |        |        |        |        |        |
| BOIN12                     | 29.9   | 38.9   | 13.5   | 6.3    | 2.8    | 8.6    |
| U-BOIN                     | 9.3    | 17.6   | 18.3   | 22.9   | 31.4   | 0.5    |
| EffTox                     | 22     | 20     | 15     | 7      | 18     | 19     |
| AIDE one_stage (Utility 2) | 20.8   | 48.7   | 15.4   | 7.8    | 1.6    | 5.7    |
| AIDE one_stage (Utility 3) | 25     | 47.6   | 13     | 7.2    | 1.2    | 6      |
| AIDE two_stage (Utility 2) | 20.4   | 40.2   | 15.9   | 6.3    | 2.3    | 14.9   |
| AIDE two_stage (Utility 3) | 24.5   | 42.5   | 11.6   | 4.9    | 1.4    | 15.1   |
| Mean patients treated      |        |        |        |        |        |        |
| BOIN12                     | 9.7    | 9.1    | 5.1    | 3.7    | 1.8    |        |
| U-BOIN                     | 8      | 7.3    | 5.6    | 5      | 4.1    |        |
| EffTox                     | 13.3   | 5.3    | 4      | 2      | 2.5    |        |
| AIDE one_stage (Utility 2) | 8.7    | 10.5   | 5.5    | 3.5    | 1.2    |        |
| AIDE one_stage (Utility 3) | 9.3    | 10.3   | 5.4    | 3.4    | 1.1    |        |
| AIDE two_stage (Utility 2) | 9.6    | 8.2    | 5.8    | 2.8    | 1.4    |        |
| AIDE two_stage (Utility 3) | 9.6    | 8.2    | 5.7    | 2.8    | 1.4    |        |
| Scenario 37                |        |        |        |        |        |        |
| DLT rate                   | 0.08   | 0.3    | 0.4    | 0.45   | 0.5    |        |
| Efficacy rate              | 0.05   | 0.2    | 0.2    | 0.2    | 0.2    |        |
| Utility 2                  | 0      | -0.1   | -0.2   | -0.2   | -0.3   |        |
| Utility 3                  | 39.8   | 40     | 36     | 34     | 32     |        |
| OBD selection (%)          |        |        |        |        |        |        |
| BOIN12                     | 39.2   | 39     | 8.9    | 1.5    | 0.5    | 10.9   |
| U-BOIN                     | 25.3   | 25.9   | 13.7   | 19.8   | 15.2   | 0.1    |
| EffTox                     | 22     | 26     | 15     | 6      | 7      | 25     |
| AIDE one_stage (Utility 2) | 18.9   | 57.1   | 16.3   | 3.8    | 0.5    | 3.4    |
| AIDE one_stage (Utility 3) | 24.7   | 55.3   | 13     | 3.1    | 0.5    | 3.4    |
| AIDE two_stage (Utility 2) | 26.2   | 49.1   | 14.1   | 3.1    | 0.4    | 7.1    |
| AIDE two_stage (Utility 3) | 32.5   | 47.7   | 10     | 2.4    | 0.3    | 7.1    |
| Mean patients treated      |        |        |        |        |        |        |
| BOIN12                     | 11.2   | 10.1   | 4.5    | 2.4    | 1.1    |        |
| U-BOIN                     | 8.7    | 8.4    | 5.3    | 4.3    | 3.4    |        |
| EffTox                     | 13.6   | 6      | 4.2    | 1.5    | 1.2    |        |
| AIDE one_stage (Utility 2) | 8.4    | 11.7   | 5.7    | 3      | 0.9    |        |
| AIDE one_stage (Utility 3) | 9.1    | 11.4   | 5.4    | 3      | 0.8    |        |
| AIDE two_stage (Utility 2) | 8.7    | 10.7   | 6.1    | 2.4    | 0.9    |        |
| AIDE two_stage (Utility 3) | 8.6    | 10.7   | 6.1    | 2.4    | 0.9    |        |